

MAGDAGACHI, Russian FederationWMO: **312950**Lat: **53.467N**Lon: **125.817E**Elev: **363**StdP: **97.04**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.4	-33.2	-37.8	0.1	-34.8	-35.9	0.1	-32.7	9.1	-18.9	8.1	-17.9	0.8	270	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	30.1	20.1	28.3	19.3	26.6	18.6	21.8	27.0	20.7	25.8	19.7	24.6	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.5	24.2	19.0	14.4	23.1	18.0	13.5	22.2	65.7	26.9	61.9	25.9	58.4	24.4	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	7.7	6.4	-38.1	33.3	2.9	1.7	-40.2	34.5	-41.9	35.6	-43.5	36.5	-45.6	37.8		
(5)			-38.1	24.1	2.9	1.8	-40.2	25.4	-41.9	26.5	-43.5	27.5	-45.5	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.0	-23.2	-18.6	-9.8	1.8	10.4	17.4	20.0	17.4	9.8	-0.7	-14.9	-23.1
	DBStd	16.34	4.96	4.86	5.82	4.76	4.09	3.52	2.88	3.31	4.28	5.27	6.29	4.91
	HDD10.0	4901	1028	802	613	251	45	1	0	1	55	333	746	1027
	HDD18.3	7194	1286	1035	872	497	248	57	16	57	256	590	996	1286
	CDD10.0	876	0	0	0	4	57	223	311	231	49	1	0	0
	CDD18.3	128	0	0	0	0	1	29	68	29	1	0	0	0
	CDH23.3	1341	0	0	0	2	39	392	611	284	13	0	0	0
	CDH26.7	353	0	0	0	0	6	107	181	59	1	0	0	0
Wind	WSAvg	2.6	2.2	2.6	3.1	3.4	3.2	2.4	2.1	2.2	2.7	2.8	2.4	2.3
Precipitation	PrecAvg	484	7	6	6	18	50	76	120	101	53	26	16	9
	PrecMax	923	16	14	15	53	107	138	356	202	111	78	38	18
	PrecMin	367	1	0	0	1	2	22	55	26	20	3	1	2
	PrecStd	109	5	4	5	12	28	33	67	45	25	19	9	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.5	-2.5	8.1	21.4	27.4	31.9	33.2	30.4	25.1	16.7	1.6	-6.6
		MCWB	-10.1	-5.1	2.5	11.1	16.4	19.6	21.5	20.6	17.1	9.6	-1.3	-8.3
	2%	DB	-11.5	-5.9	4.9	16.4	24.1	29.7	30.6	28.4	22.5	12.7	-1.3	-10.0
		MCWB	-12.7	-7.8	0.3	8.0	13.7	18.9	21.0	19.5	15.6	7.5	-3.2	-11.2
	5%	DB	-13.6	-8.4	2.0	13.0	21.5	27.6	28.6	26.5	20.1	9.6	-3.1	-13.6
		MCWB	-14.6	-10.0	-1.8	6.0	12.5	18.0	20.2	19.0	13.9	5.2	-4.5	-14.6
	10%	DB	-15.6	-10.8	-0.5	10.1	19.0	25.6	26.8	24.6	17.6	7.0	-5.5	-16.0
		MCWB	-16.4	-12.2	-3.5	4.3	11.1	17.2	19.4	18.2	12.5	3.3	-6.8	-16.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.0	-5.1	2.7	11.3	16.8	21.8	24.2	22.5	18.4	10.5	-1.1	-8.3
		MCDB	-8.8	-2.6	7.7	20.7	26.4	28.3	29.9	27.6	23.4	15.8	0.8	-6.7
	2%	WB	-12.7	-7.9	0.5	8.5	14.7	20.2	22.4	21.1	16.5	7.6	-3.0	-11.1
		MCDB	-11.6	-6.0	4.7	15.8	22.2	27.0	27.9	25.8	21.0	12.3	-1.6	-9.9
	5%	WB	-14.6	-10.0	-1.6	6.3	13.2	19.0	21.3	20.1	14.8	5.4	-4.6	-14.6
		MCDB	-13.7	-8.5	1.6	12.7	19.8	25.7	26.5	24.4	18.6	9.2	-3.1	-13.7
	10%	WB	-16.4	-12.2	-3.4	4.5	11.7	17.9	20.3	19.0	13.3	3.5	-6.8	-16.8
		MCDB	-15.6	-10.8	-0.6	9.7	17.8	24.1	25.1	23.3	16.9	6.7	-5.6	-16.1
Mean Daily Temperature Range	5% DB	MDBR	10.1	12.0	12.0	11.4	12.5	12.4	10.8	11.5	11.9	10.1	9.8	9.0
		MCDBR	10.4	13.5	13.0	16.4	17.3	15.9	14.6	14.9	15.7	13.7	10.8	9.5
		MCWBR	9.4	11.7	9.6	10.3	9.6	7.4	7.1	7.4	9.2	9.3	8.8	8.7
	5% WB	MCDBR	10.4	13.4	12.7	15.5	15.2	13.6	11.7	11.6	12.9	13.1	10.5	9.5
		MCWBR	9.4	11.7	9.5	10.0	9.1	7.4	6.8	6.8	8.3	9.2	8.7	8.7
Clear-Sky Solar Irradiance	taub	0.248	0.279	0.337	0.501	0.547	0.501	0.471	0.408	0.374	0.349	0.285	0.250	
	taud	2.152	2.110	2.010	1.759	1.766	1.984	2.128	2.297	2.317	2.211	2.141	2.091	
	Ebn at Noon	724	811	826	718	711	758	777	811	792	714	651	627	
	Edh at Noon	65	96	133	198	208	169	144	115	99	88	66	57	
All-Sky Solar Radiation	RadAvg	1.26	2.42	4.18	5.27	5.27	6.10	5.28	4.65	3.54	2.27	1.38	0.95	
	RadStd	0.05	0.07	0.15	0.46	0.35	0.55	0.54	0.39	0.28	0.15	0.08	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page